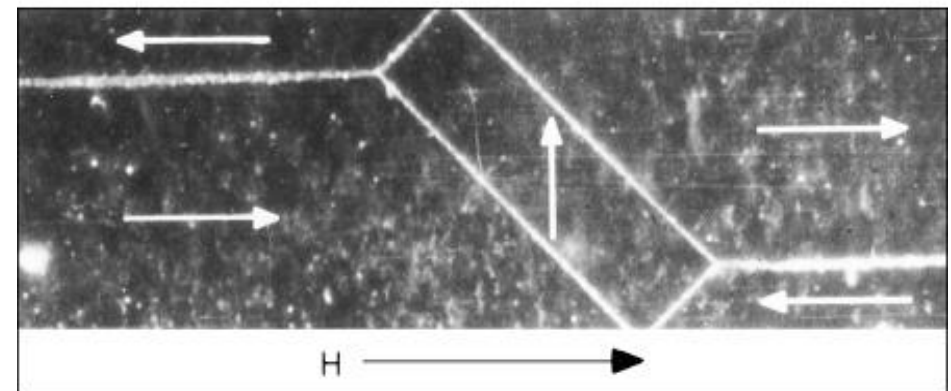
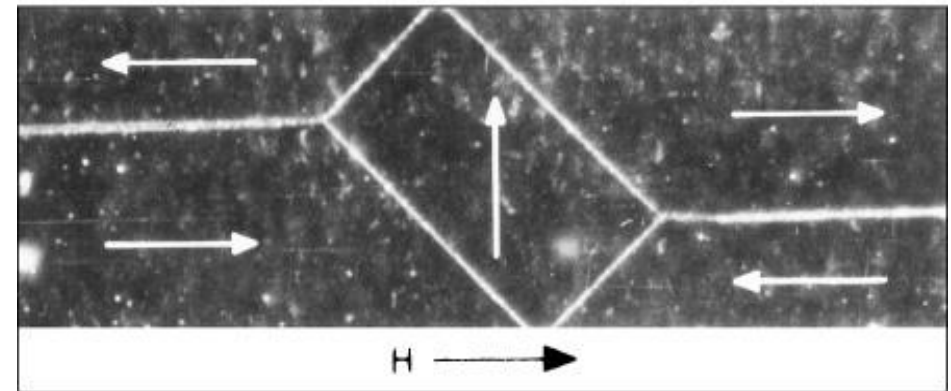
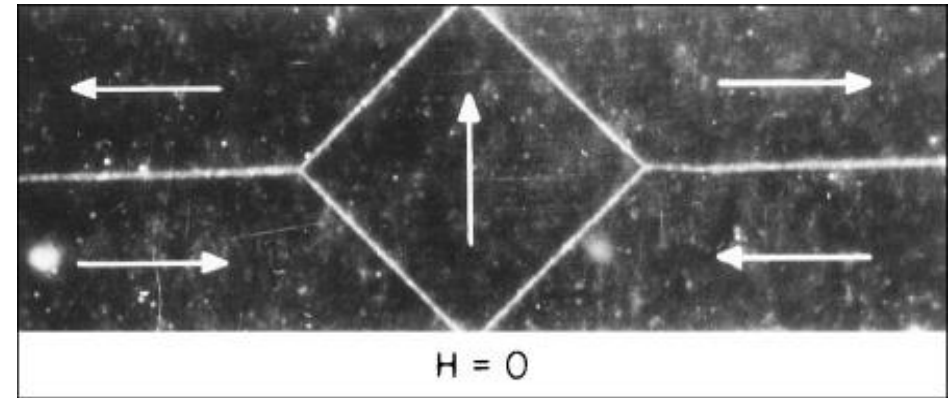


Chapter 12 Magnetic Properties

Iron single crystal photomicrographs

Magnetic domains change shape as a magnetic field (H) is applied.

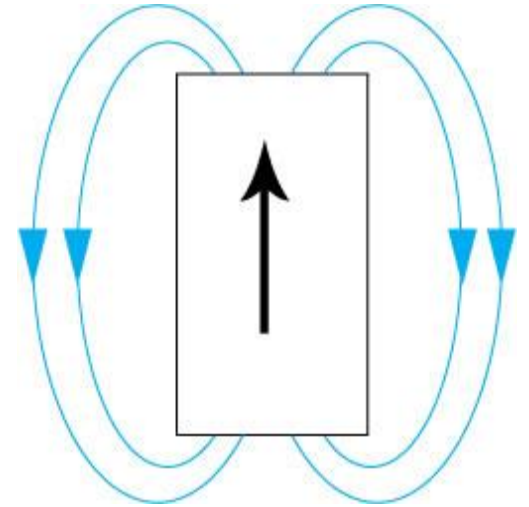
domains favorably oriented with the field grow at the expense of the unfavorably oriented domains.



Magnetic Ceramics

Aims:

- Understand the basic concepts of Related to Magnetism;
- Be aware that magnetism is affected by temperature;
- Understand the factors contributing to the formation of magnetic domains;
- Know why hysteresis occurs, and the factors which affect it.



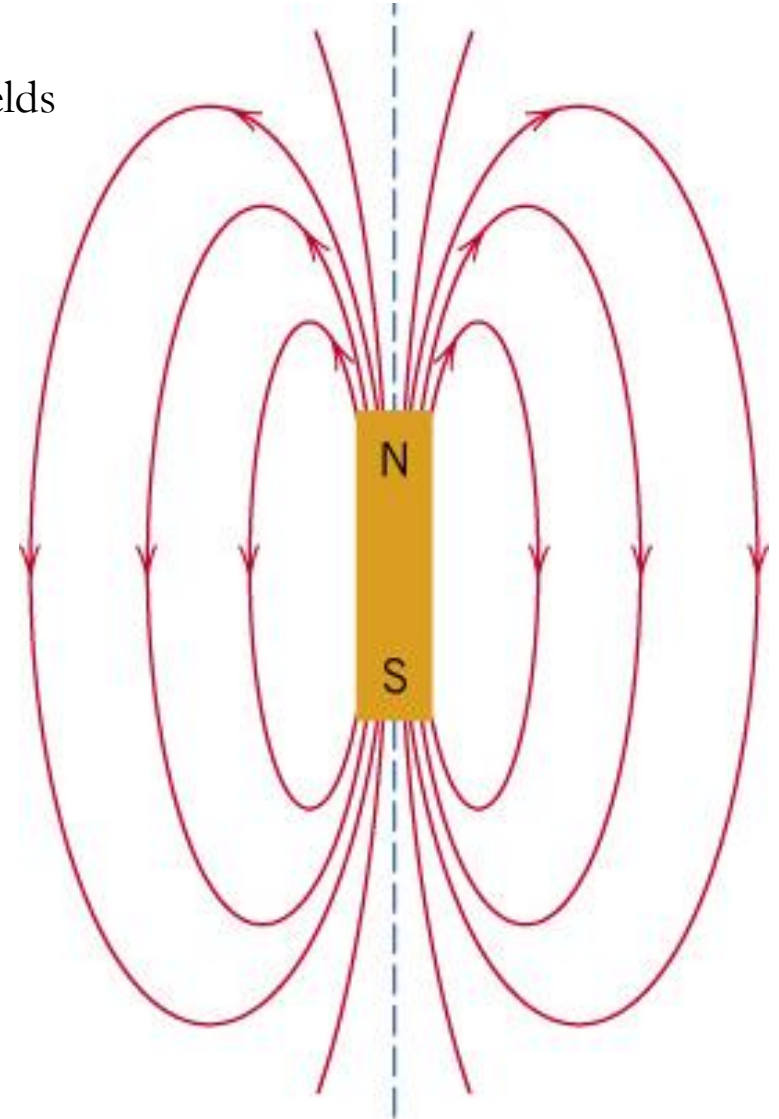
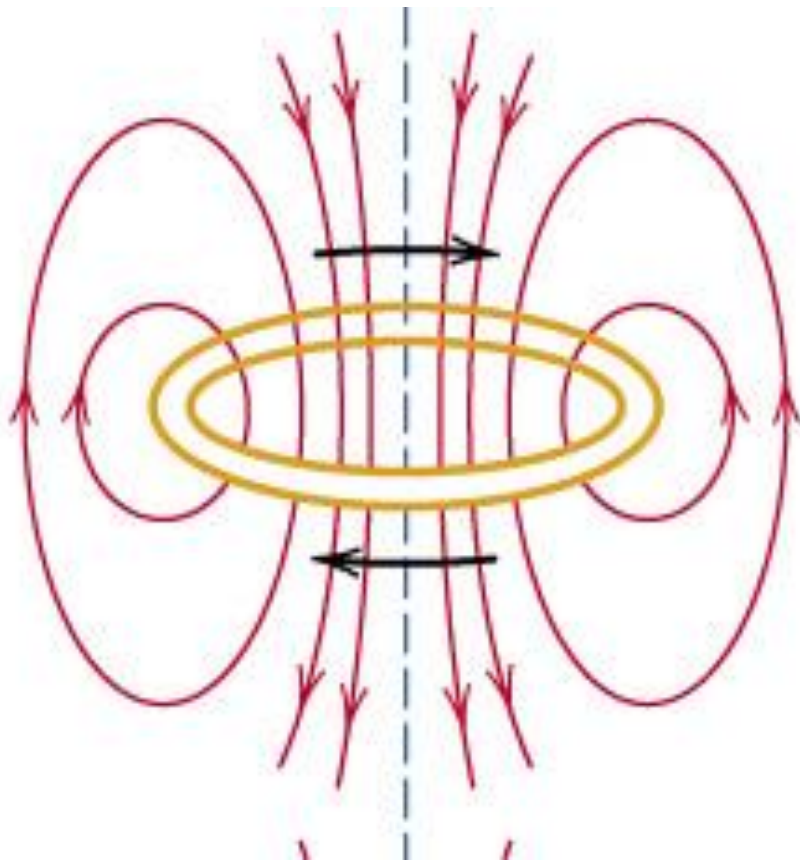
Magnetic ceramics, oxide materials that exhibit a certain type of permanent magnetization.

Basic Concepts

Magnetic fields are generated by moving electrically charged particles

Magnetic forces appear when moving charges

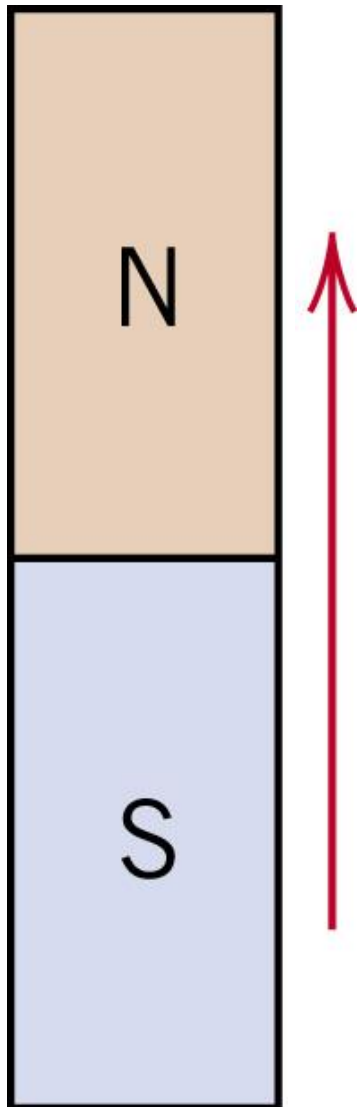
Forces can be represented by imaginary lines grouped as fields



Magnetic field lines of force around a current loop and a bar magnet.

Magnetic Dipoles (磁偶极子)

The magnetic moment represented by a **vector**



Magnetic dipoles are found to exist in magnetic materials, which, in some respects, are analogous to electric dipoles. **Magnetic dipoles** may be thought of as small bar magnets composed of north and south poles instead of positive and negative electric charges.

Magnetic dipoles are influenced by magnetic fields in a manner **similar to** the way in which **electric dipoles** are affected by electric fields.

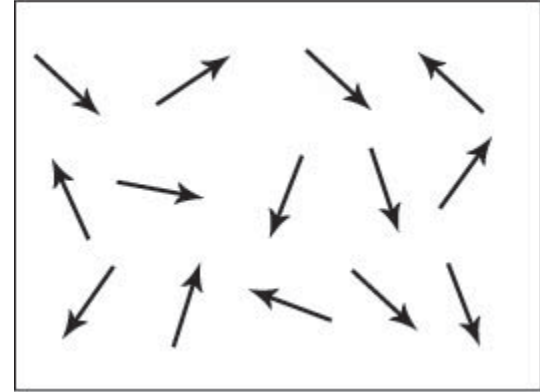
Within a magnetic field, the force of the field itself exerts a torque that tends to orient the dipoles with the field.

A familiar example is the way in which a magnetic compass needle lines up with the earth's magnetic field.

Types of Magnetism

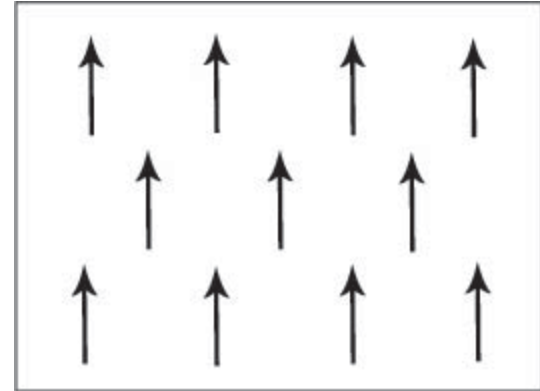
Paramagnetism (顺磁性)

In a paramagnet, the **magnetic moments** tend to be **randomly orientated** due to thermal fluctuations when there is **no magnetic field**. In an applied magnetic field these moments start to align parallel to the field such that the magnetization of the material is proportional to the applied field. Paramagnets **do not retain any magnetization** in the absence of an externally applied magnetic field because thermal motion randomizes the spin orientations.



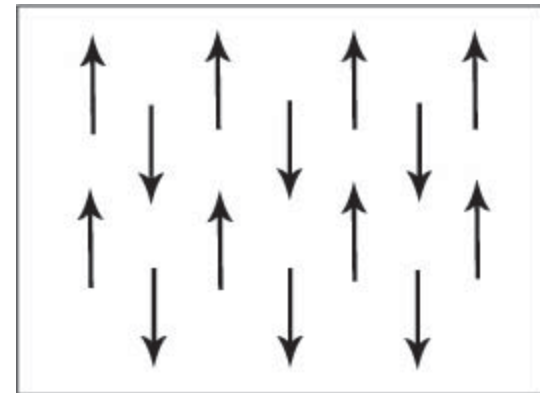
Ferromagnetism (铁磁性)

The magnetic moments in a ferromagnet have the tendency to become aligned parallel to each other under the influence of a magnetic field. However, unlike the moments in a paramagnet, **these moments will then remain parallel** when a **magnetic field is not applied**.



Antiferromagnetism (反铁磁性)

Adjacent magnetic moments from the magnetic ions tend to align anti-parallel to each other without an applied field. In the simplest case, adjacent magnetic moments are equal in magnitude and opposite therefore there is no overall magnetization.

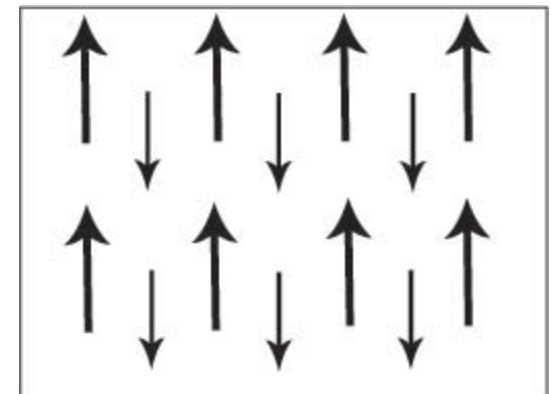


Ferrimagnetism (铁氧体)

Ferrimagnetism

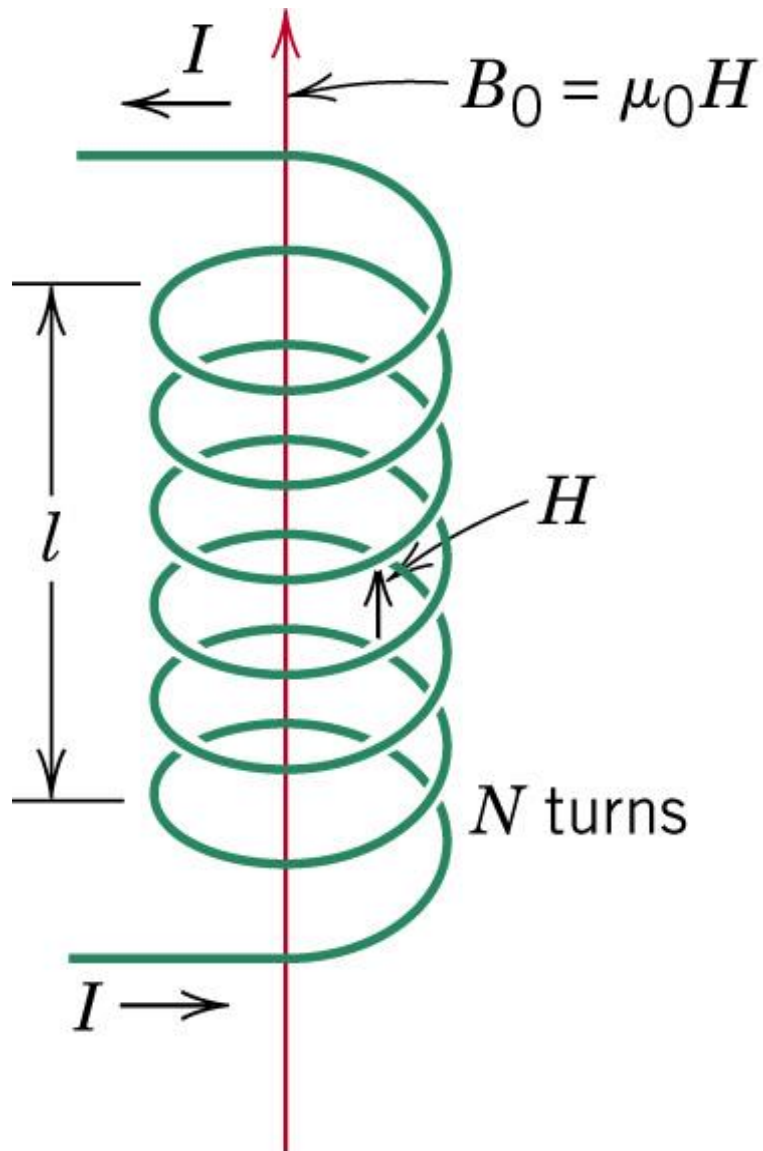
The aligned magnetic moments are not of the same size; that is to say there is more than one type of magnetic ion. An overall magnetization is produced but **not all the magnetic moments may give a positive contribution** to the overall magnetization.

Diagram of a periodic table showing elements colored according to the type of magnetism they show at room temperature.

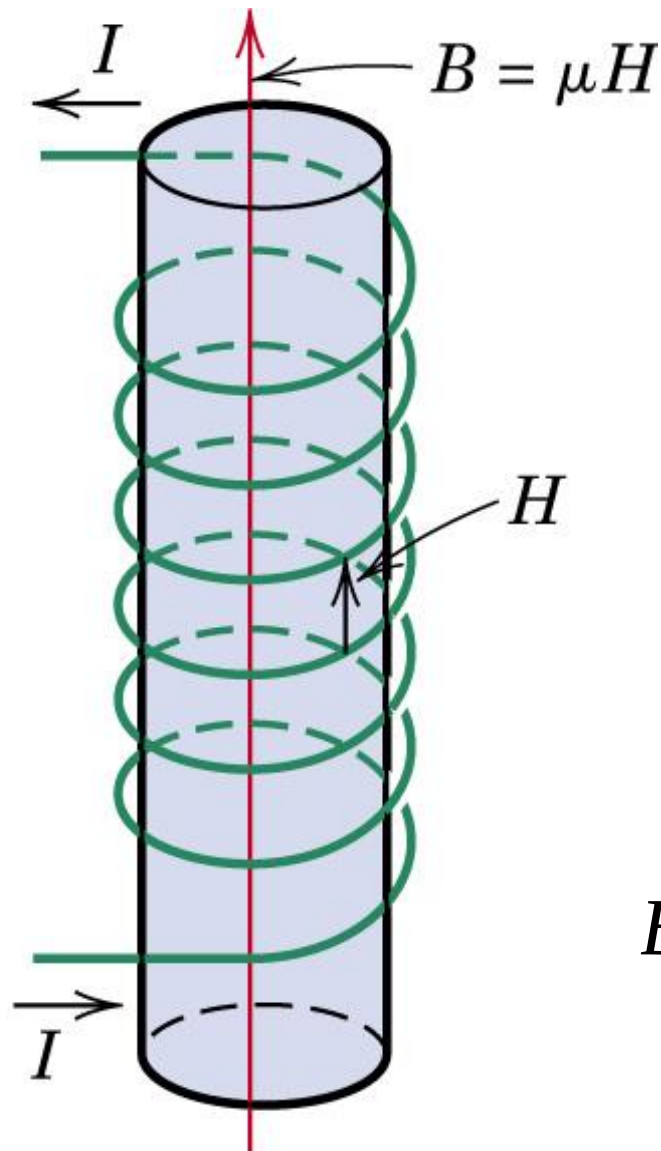
[illegible]

Magnetic Field Vectors

Magnetic field strength (H) & magnetic flux density (B)



Magnetic field strength $H = \frac{NI}{l}$



Magnetic flux density

$$B = \mu H$$

$$B_0 = \mu_0 H$$

Relative permeability
相对磁导率

$$\mu_r = \frac{\mu}{\mu_0}$$

magnetization

$$B = \mu_0 H + \mu_0 M$$

$$M = \chi_m H$$

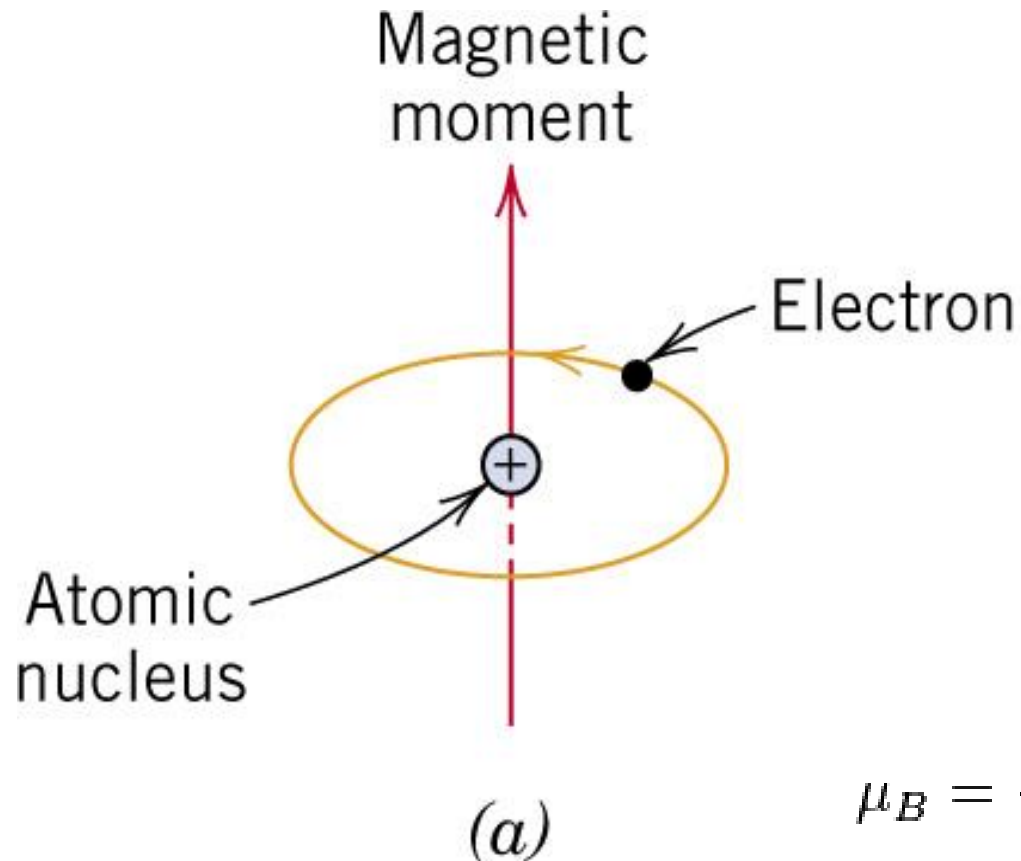
Magnetic susceptibility
磁化率

$$\chi_m = \mu_r - 1$$

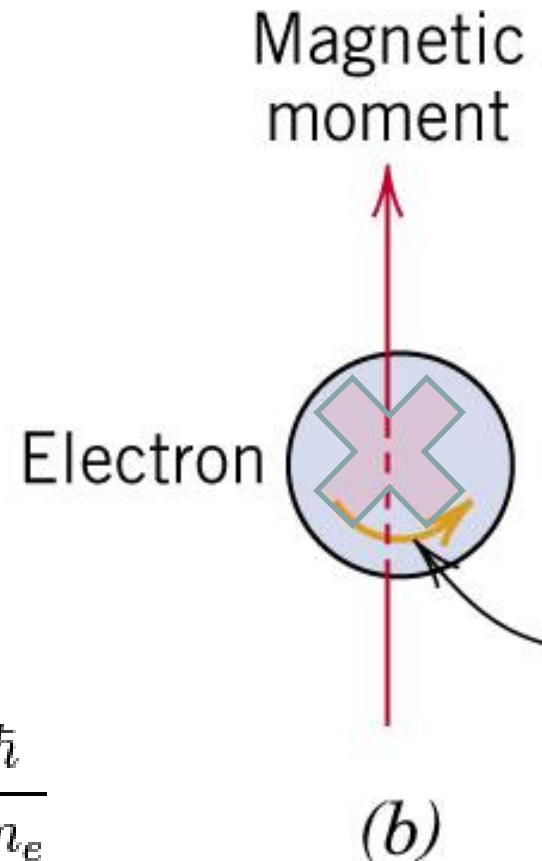
Origins of Magnetic Moments

Responds to quantum mechanics laws

Two main contributions: *(a) an orbiting electron and (b) electron spin.*



$$\mu_B = \frac{e\hbar}{2m_e}$$



The spin is an intrinsic property of the electron and it is not due to its rotation

Bohr magneton (m_B)

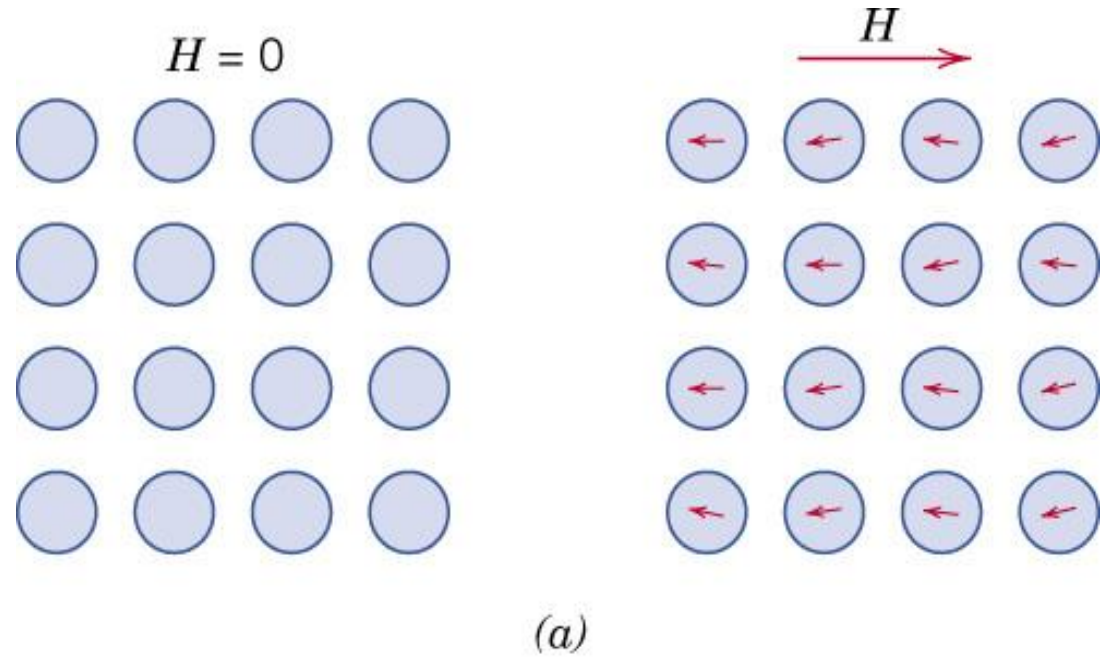
Most fundamental magnetic moment

$$m_B = \pm 9.27 \times 10^{-24} \text{ A}\cdot\text{m}^2$$

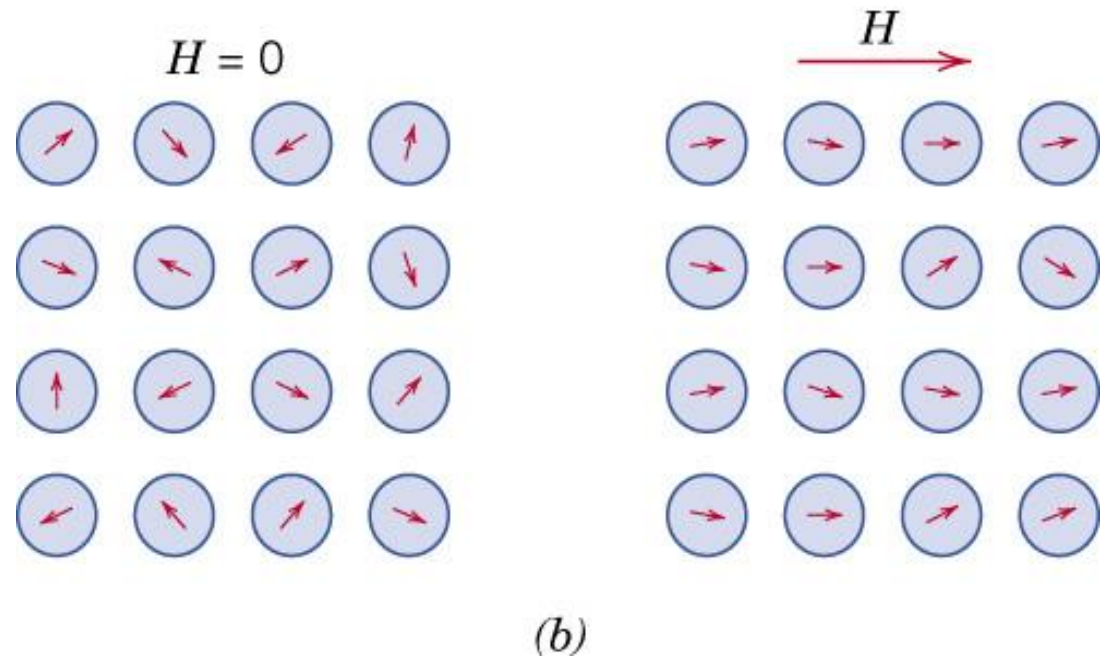
Diamagnetism and Paramagnetism

Diamagnetic material

in the presence of a field, dipoles are induced and aligned opposite to the field direction.



Paramagnetic material



$$B = \mu_0 H + \mu_0 M = \mu_0 H + \mu_0 \chi_m H$$

$$\mu = \mu_0 (1 + \chi_m)$$

The flux density B versus the magnetic field strength H for diamagnetic and paramagnetic materials.

$$\mu_r = \frac{\mu}{\mu_0}$$

$$M = \chi_m H$$

$$\chi_m = \mu_r - 1$$

Paramagnetic

Vacuum

Diamagnetic

Flux density, B

$$B = \mu H$$

$$B_0 = \mu_0 H$$

0
0

Magnetic field strength, H

Comparison of Room-Temperature Magnetic Susceptibilities (磁化系数)

Room-Temperature Magnetic Susceptibilities for Diamagnetic and Paramagnetic Materials

<i>Diamagnetics</i>		<i>Paramagnetics</i>	
<i>Material</i>	<i>Susceptibility χ_m (volume) (SI units)</i>	<i>Material</i>	<i>Susceptibility χ_m (volume) (SI units)</i>
Aluminum oxide	-1.81×10^{-5}	Aluminum	2.07×10^{-5}
Copper	-0.96×10^{-5}	Chromium	3.13×10^{-4}
Gold	-3.44×10^{-5}	Chromium chloride	1.51×10^{-3}
Mercury	-2.85×10^{-5}	Manganese sulfate	3.70×10^{-3}
Silicon	-0.41×10^{-5}	Molybdenum	1.19×10^{-4}
Silver	-2.38×10^{-5}	Sodium	8.48×10^{-6}
Sodium chloride	-1.41×10^{-5}	Titanium	1.81×10^{-4}
Zinc	-1.56×10^{-5}	Zirconium	1.09×10^{-4}

FERROMAGNETISM

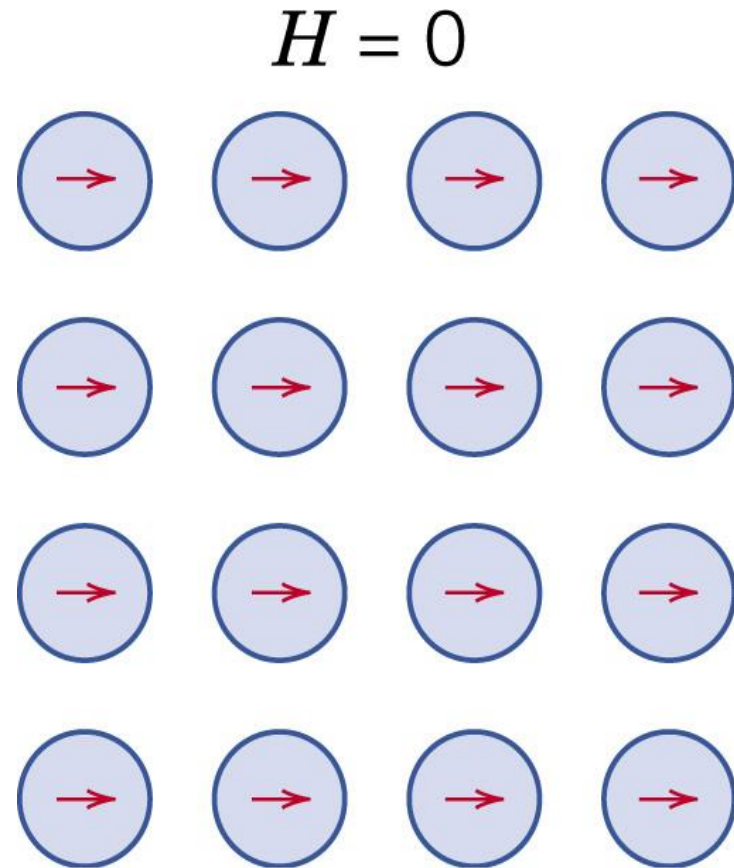
Mutual alignment of atomic dipoles

even in the absence of an external magnetic field.

coupling forces align the magnetic spins

$$B = \mu_0 H + \mu_0 M$$

$$B \approx \mu_0 M$$



Domains with mutual spin alignment

B grows up to a saturation magnetization M_s with a saturation flux

$B_s = M_{\text{atom}} \times N_{\text{atoms}}$ (average moment per atom times density of atoms)

$M_{\text{atom}} = 2.22m_B, 1.72m_B, 0.60m_B$ for Fe, Co, Ni, respectively

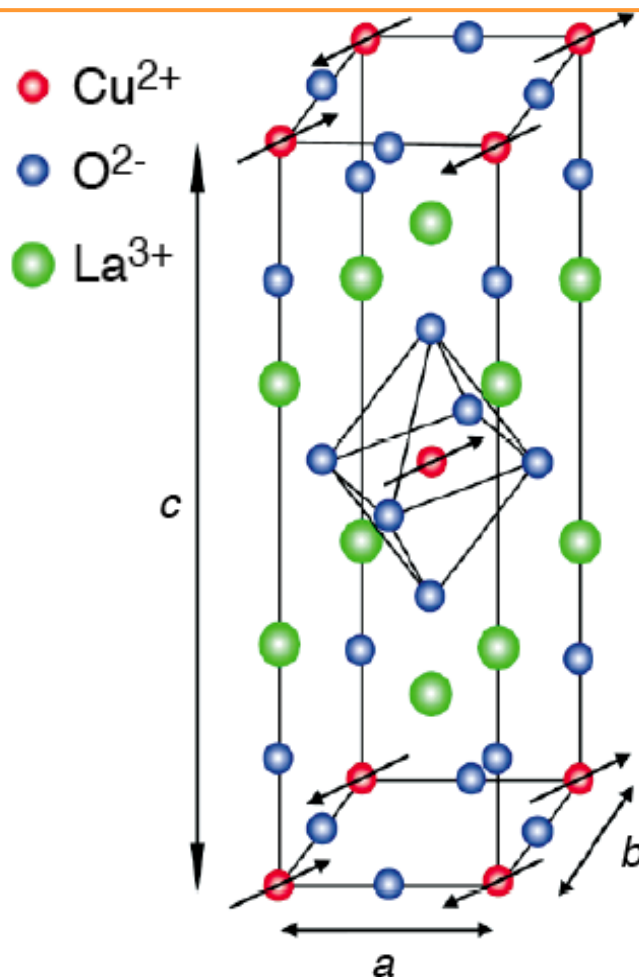
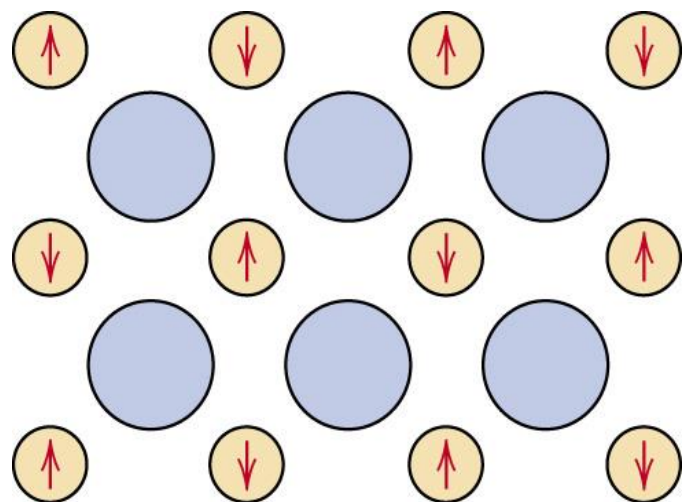
Antiferromagnetism & Ferrimagnetism

ANTIFERROMAGNETISM

Antiparallel alignment of spin magnetic moments for antiferromagnetic manganese oxide (MnO)

At low T

Above the Neel temperature they become paramagnetic



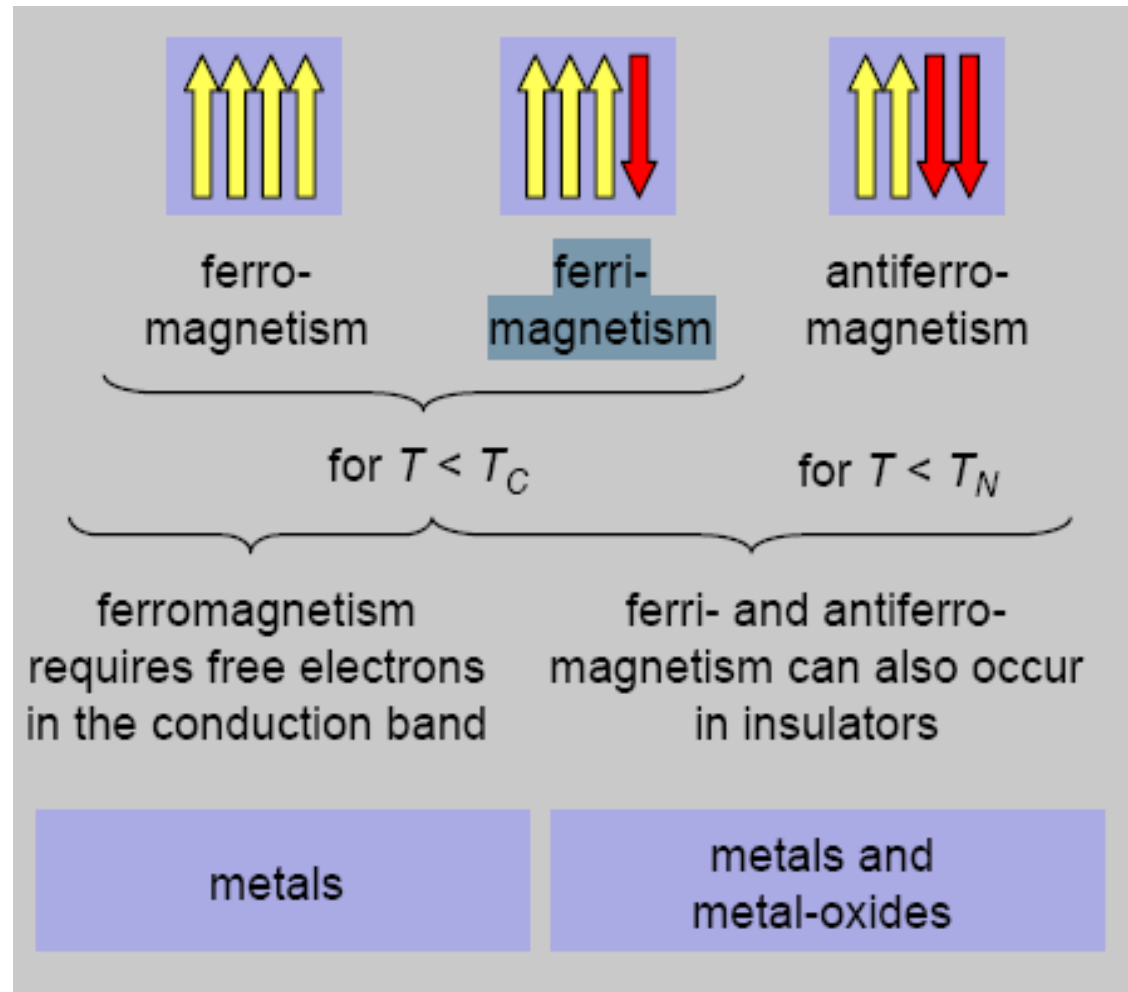
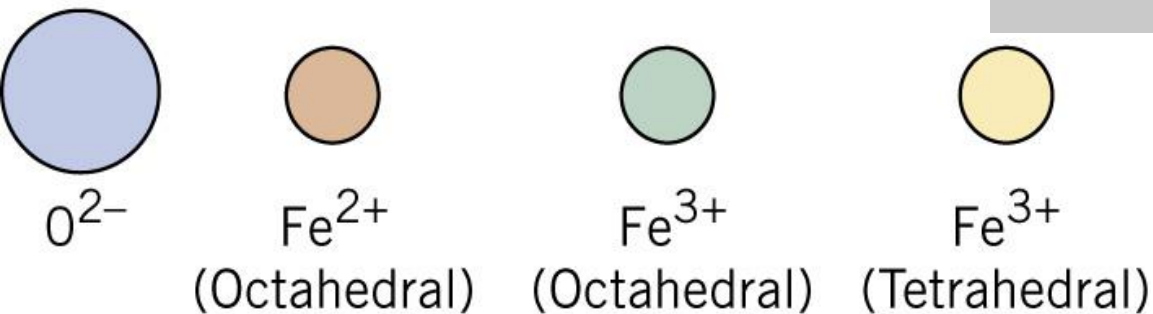
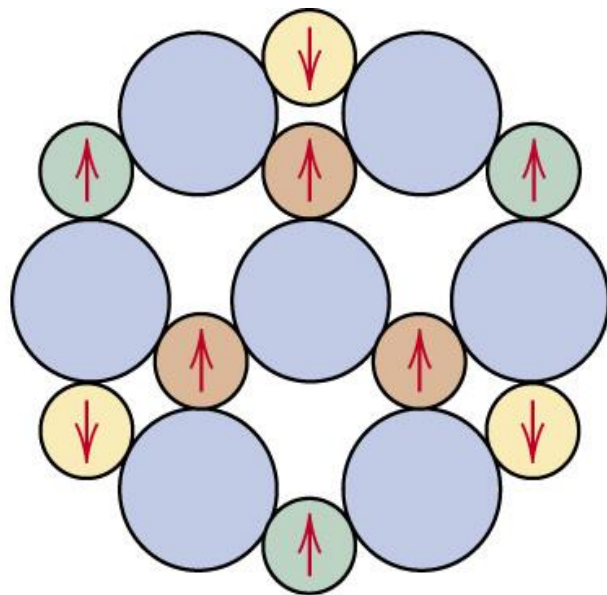
1986: superconductivity discovered in layered compound $\text{La}_{2-x}\text{Ba}_x\text{CuO}_4$ with a transition T much higher than expected. Little was known about copper oxides

Parent materials, La_2CuO_4 , and $\text{YBa}_2\text{Cu}_3\text{O}_6$, demonstrated that the CuO_2 planes exhibit antiferromagnetic order.

This work initiated a continuing exploration of magnetic excitations in copper-oxide superconductors, crucial to the mechanism of high-temperature superconductivity.

Ferrimagnetism (亚铁磁)

Spin magnetic moment configuration for Fe^{2+} and Fe^{3+} ions in Fe_3O_4 . Above the Curie temperature becomes paramagnetic

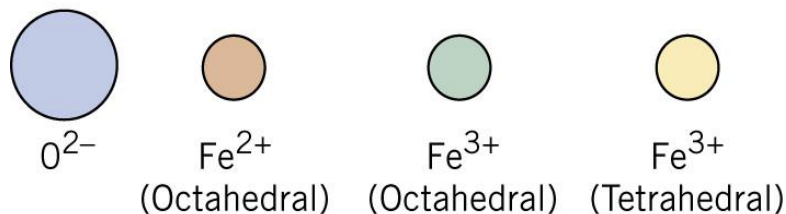
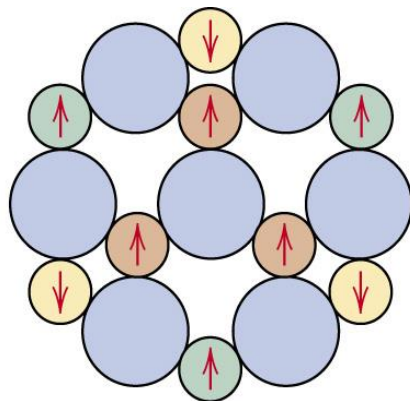


Magnetic Moments

The distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} ions in a unit cell of Fe_3O_4

<i>Cation</i>	<i>Octahedral Lattice Site</i>	<i>Tetrahedral Lattice Site</i>	<i>Net Magnetic Moment</i>
Fe^{3+}	$\uparrow \uparrow \uparrow \uparrow$ $\uparrow \uparrow \uparrow \uparrow$	$\downarrow \downarrow \downarrow \downarrow$ $\downarrow \downarrow \downarrow \downarrow$	Complete cancellation
Fe^{2+}	$\uparrow \uparrow \uparrow \uparrow$ $\uparrow \uparrow \uparrow \uparrow$	—	$\uparrow \uparrow \uparrow \uparrow$ $\uparrow \uparrow \uparrow \uparrow$

^a Each arrow represents the magnetic moment orientation for one of the cations.



Net Magnetic Moments for Six Cations

<i>Cation</i>	<i>Net Spin Magnetic Moment (Bohr magnetons)</i>
Fe^{3+}	5
Fe^{2+}	4
Mn^{2+}	5
Co^{2+}	3
Ni^{2+}	2
Cu^{2+}	1

Magnetic Properties of Several Ferrimagnetic Ceramics

Material	θ_c (K)	$B_{sat}(T)$ @ RT	Calculated moments			Experimental
			T site	O site	Net	
<i>Spinel ferrites</i> [AO•B ₂ O ₃]						
Fe ³⁺ [Cu ²⁺ Fe ³⁺]O ₄	728	0.20	−5	1.73 + 5	1	1.30
Fe ³⁺ [Ni ²⁺ Fe ³⁺]O ₄	858	0.34	−5	2 + 5	2	2.40
Fe ³⁺ [Co ²⁺ Fe ³⁺]O ₄	1,020	0.50	−5	3 + 5	3	3.70–3.90
Fe ³⁺ [Fe ²⁺ Fe ³⁺]O ₄	858	0.60	−5	4 + 5	4	4.10
Fe ³⁺ [Mn ²⁺ Fe ³⁺]O ₄	573	0.51	−5	5 + 5	5	4.60–5.0
Fe ³⁺ [Li _{0.5} Fe _{1.5}]O ₄	943		−5	0 + 0.75		2.60
Mg _{0.1} Fe _{0.9} [Mg _{0.9} Fe _{1.1}]O ₄	713	0.14	0–4.5	0 + 5.5	1	1.10
<i>Hexagonal ferrites</i>						
BaO:6Fe ₂ O ₃	723	0.48				1.10
SrO:6Fe ₂ O ₃	723	0.48				1.10
Y ₂ O ₃ :5Fe ₂ O ₃	560	0.16				5.00
BaO:9Fe ₂ O ₃	718	0.65				
<i>Garnets</i>						
YIG{Y ₃ }[Fe ₂]Fe ₃ O ₁₂	560	0.16			5	4.96
(Gd ₃)[Fe ₂]Fe ₃ O ₁₂	560				16	15.20
<i>Binary oxides</i>						
EuO	69					6.8
CrO ₂	386	0.49				2.00

magnetic susceptibility χ_m

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H (-2,5)	all values given for a temperature of 300 K in case of ferromagnetic materials: saturation polarization																He (-1,1)
Li 24	Be -23											B -19	C -22	N (-6,3)	O 7,9	F	Ne (-4,0)
Na 8,1	Mg 5,7											Al 21	Si -3,4	P -23	S -12	Cl (-22)	Ar (-11)
K 5,7	Ca 21	Sc 264	Ti 181	V 383	Cr 267	Mn 828	Fe 2,16	Co 1,76	Ni 0,61	Cu -9,7	Zn -12	Ga -23	Ge -7,3	As -5,4	Se -18	Br -16	Kr (-16)
Rb 4,4	Sr 36	Y 122	Zr 109	Nb 236	Mo 119	Tc 373	Ru 66	Rh 170	Pd 783	Ag -25	Cd -19	In -8,2	Sn 2,4	Sb -67	Te -24	I -22	Xe (-24)
Cs 5,3	Ba 6,7	La 63	Hf 71	Ta 175	W 78	Re 96	Os 15	Ir 37	Pt 264	Au -34	Hg -28	Tl -36	Pb -16	Bi -153	Po	At	Rn

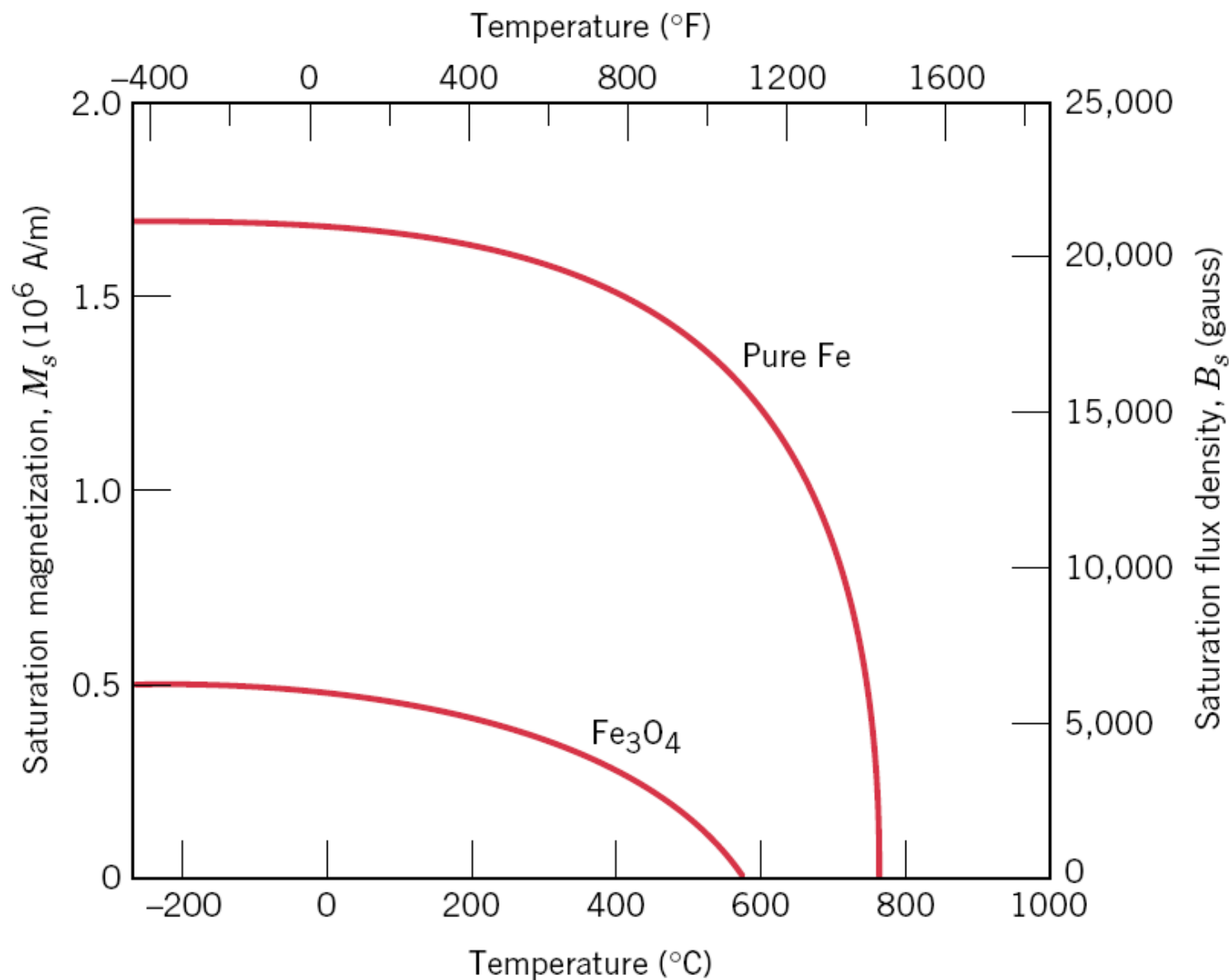
diamagnetic

paramagnetic

ferromagnetic

numbers without (): $\cdot 10^{-6}$
numbers with (): $\cdot 10^{-9}$

The Influence of Temperature on Magnetic Behavior

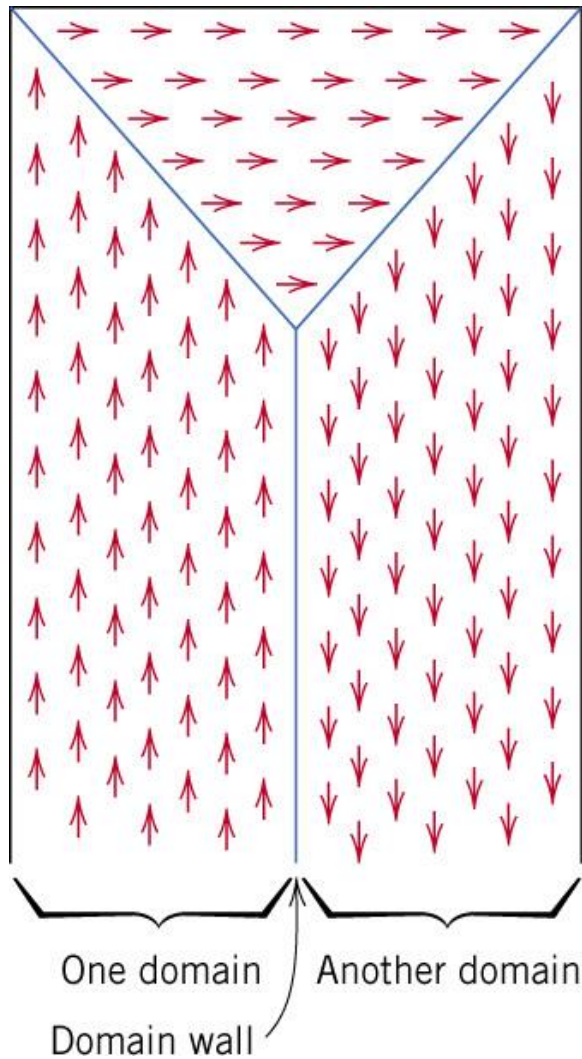


T_C : Curie temperature (ferromagnetic, ferrimagnetic); T_N : Neel temperature (antiferromagnetic), Material become paramagnetic

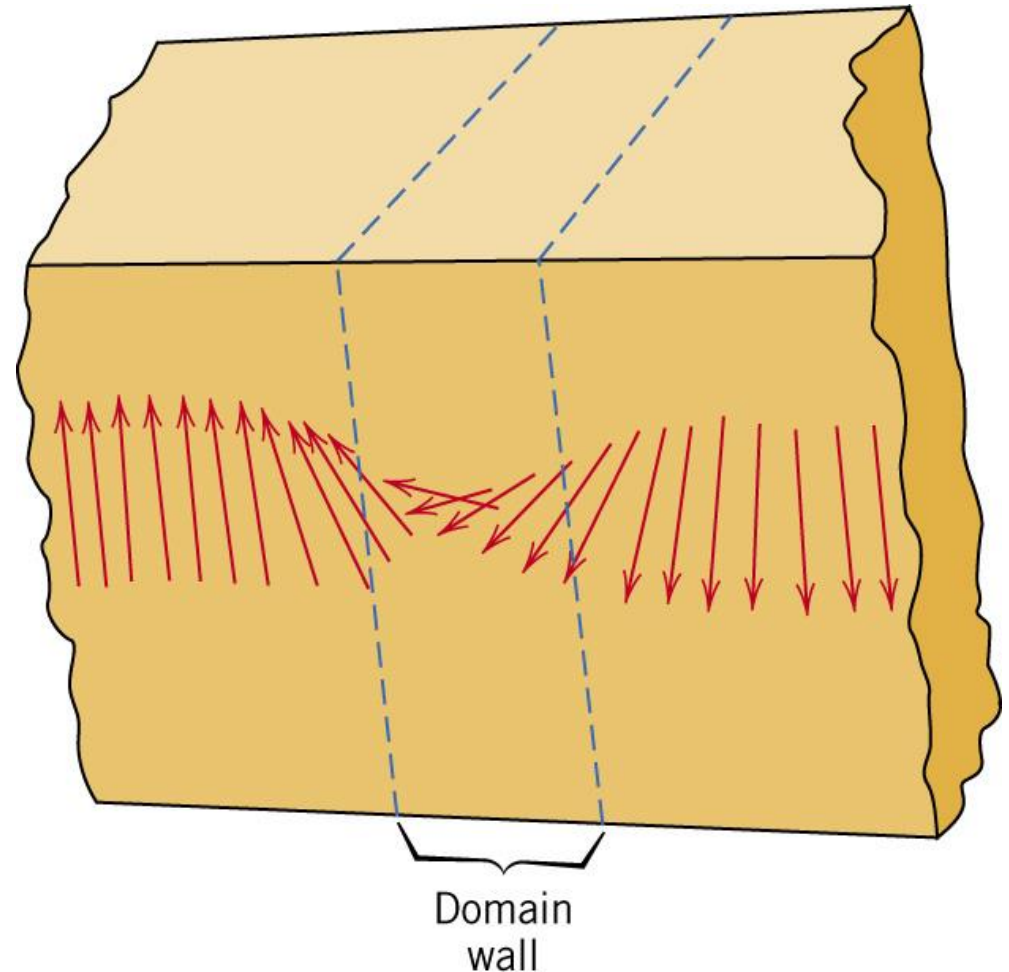
Domains and Hysteresis

Domains in a ferromagnetic or ferrimagnetic material; arrows represent atomic magnetic dipoles.

Within each domain, all dipoles are aligned, whereas the direction of alignment varies from one domain to another.



Gradual change in magnetic dipole orientation across a domain wall.

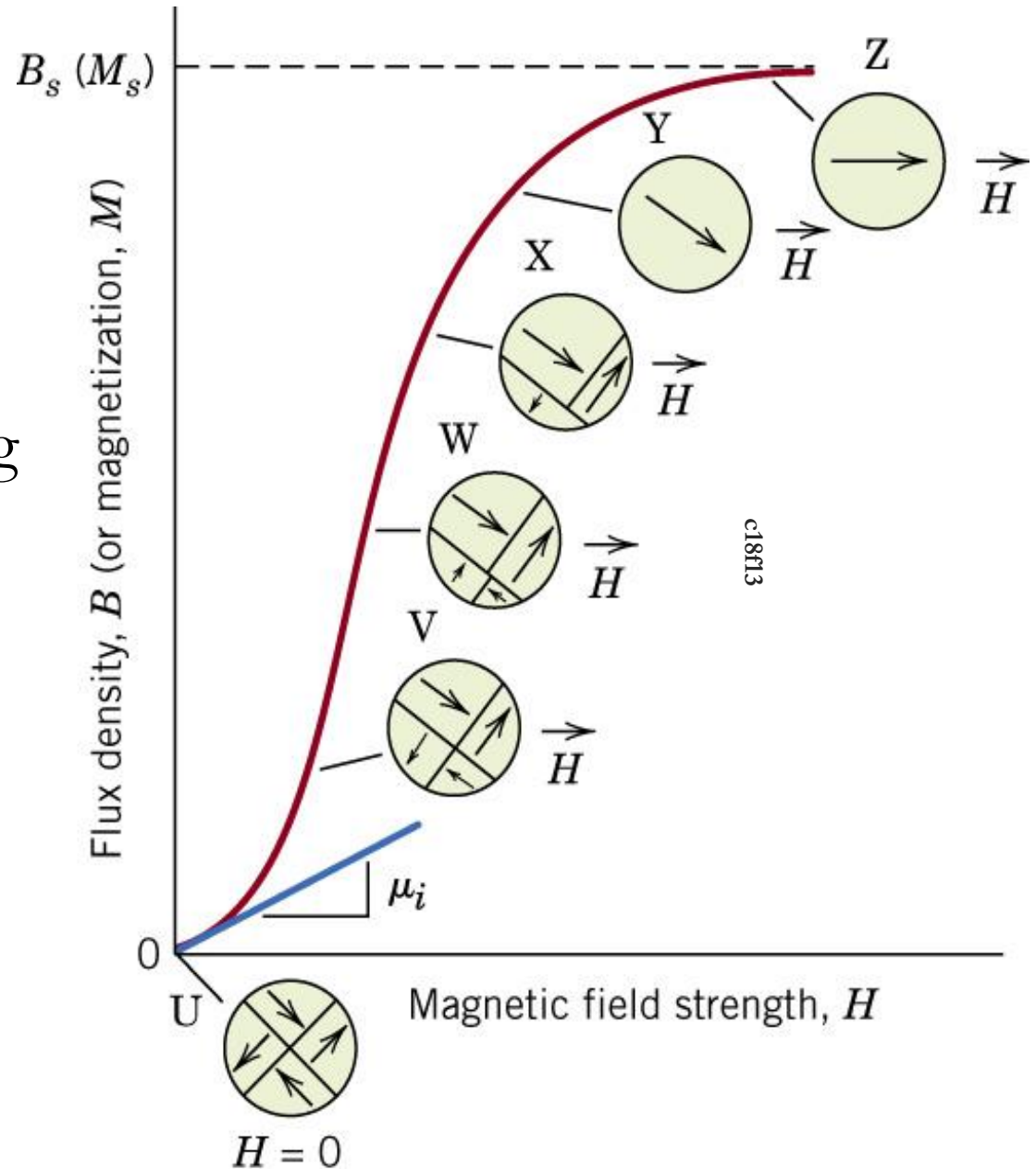


B versus H

ferromagnetic or
ferrimagnetic material
initially unmagnetized

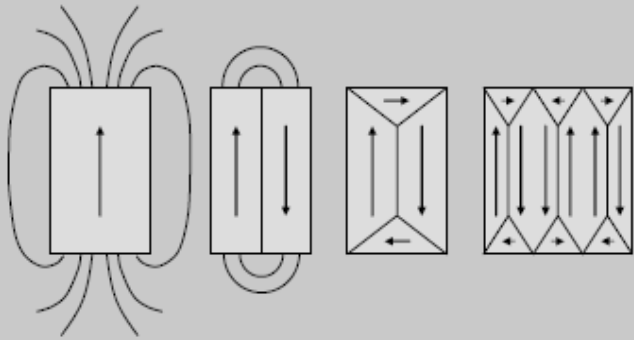
Domain configurations during
several stages of
magnetization

Saturation flux density, B_s
Magnetization, M_s
initial permeability μ_i



Magnetic flux density versus magnetic field strength

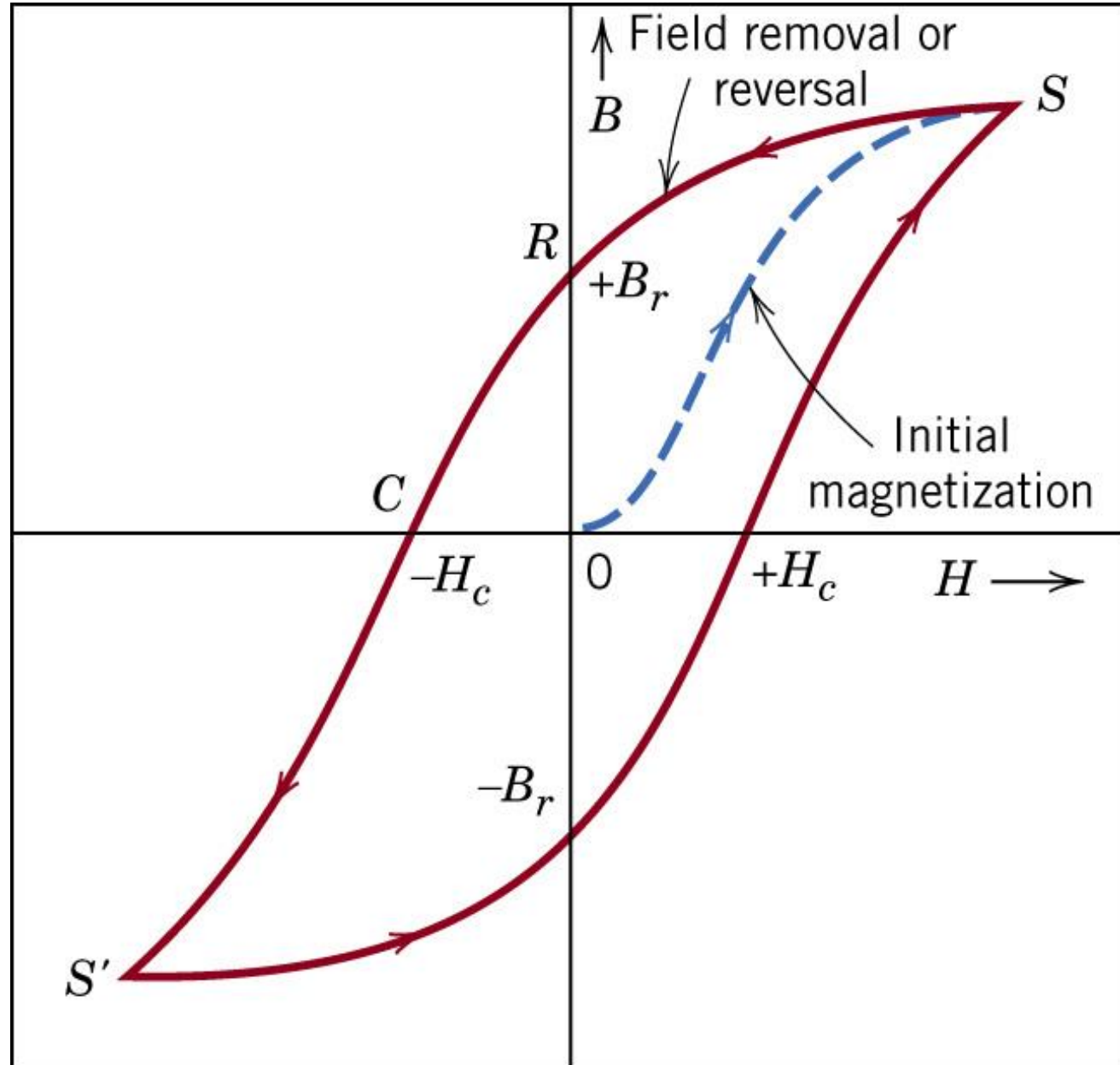
domain formation



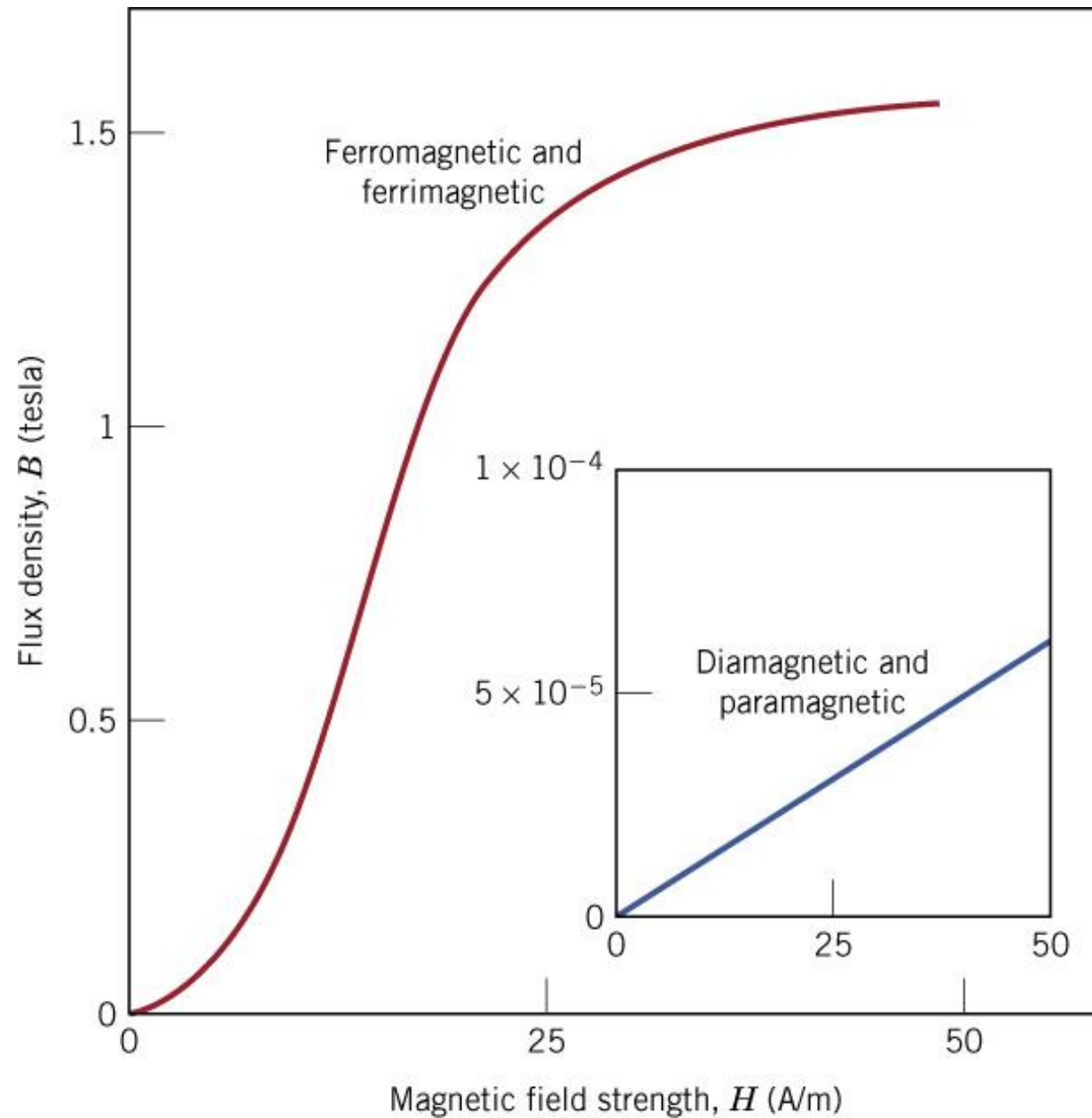
ferromagnetic material subjected to forward and reverse saturations (S & S').

hysteresis loop (red)
initial magnetization (blue)

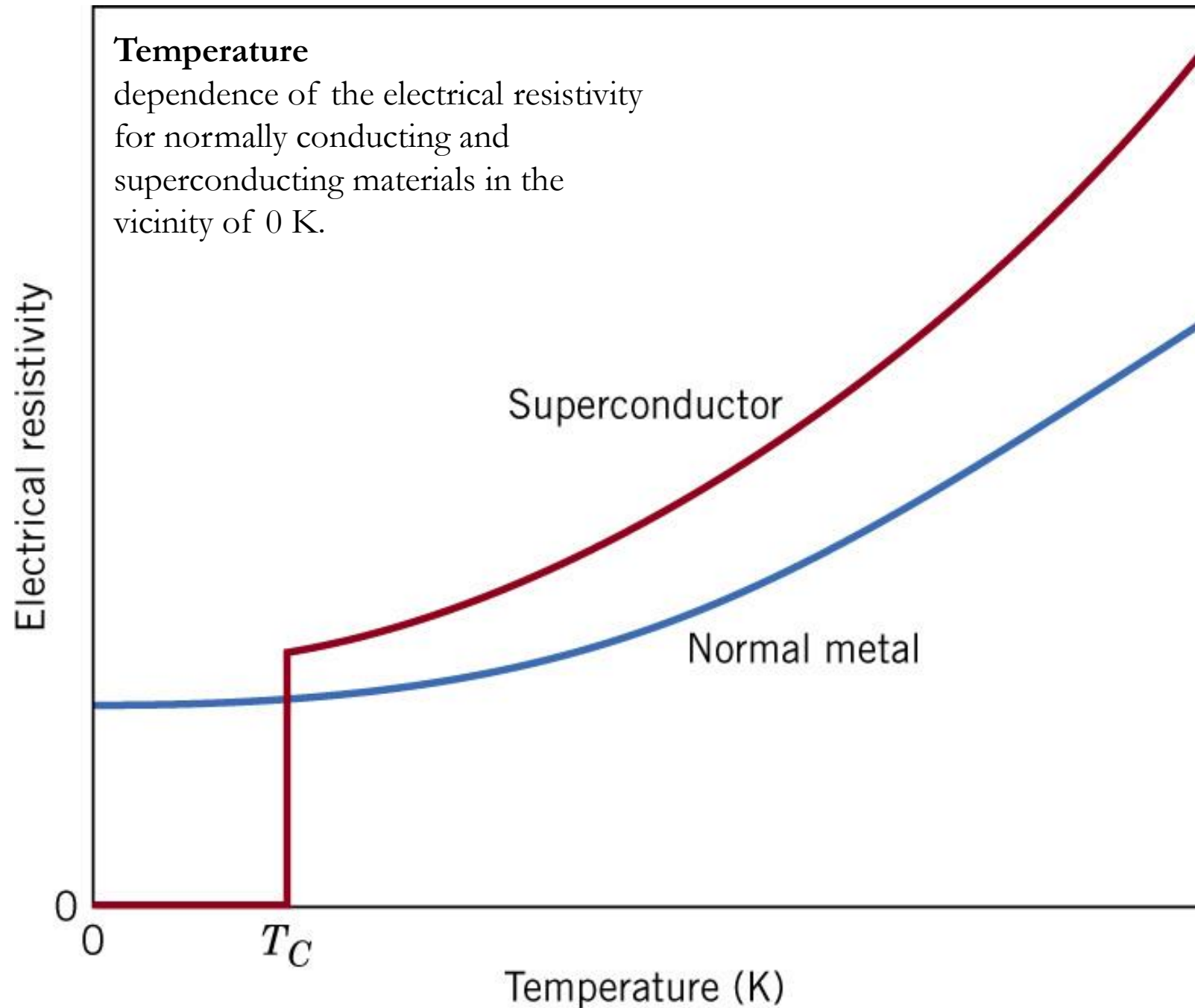
remanence, B_r
coercive force, H_c



Comparison magnetic versus nonmagnetic

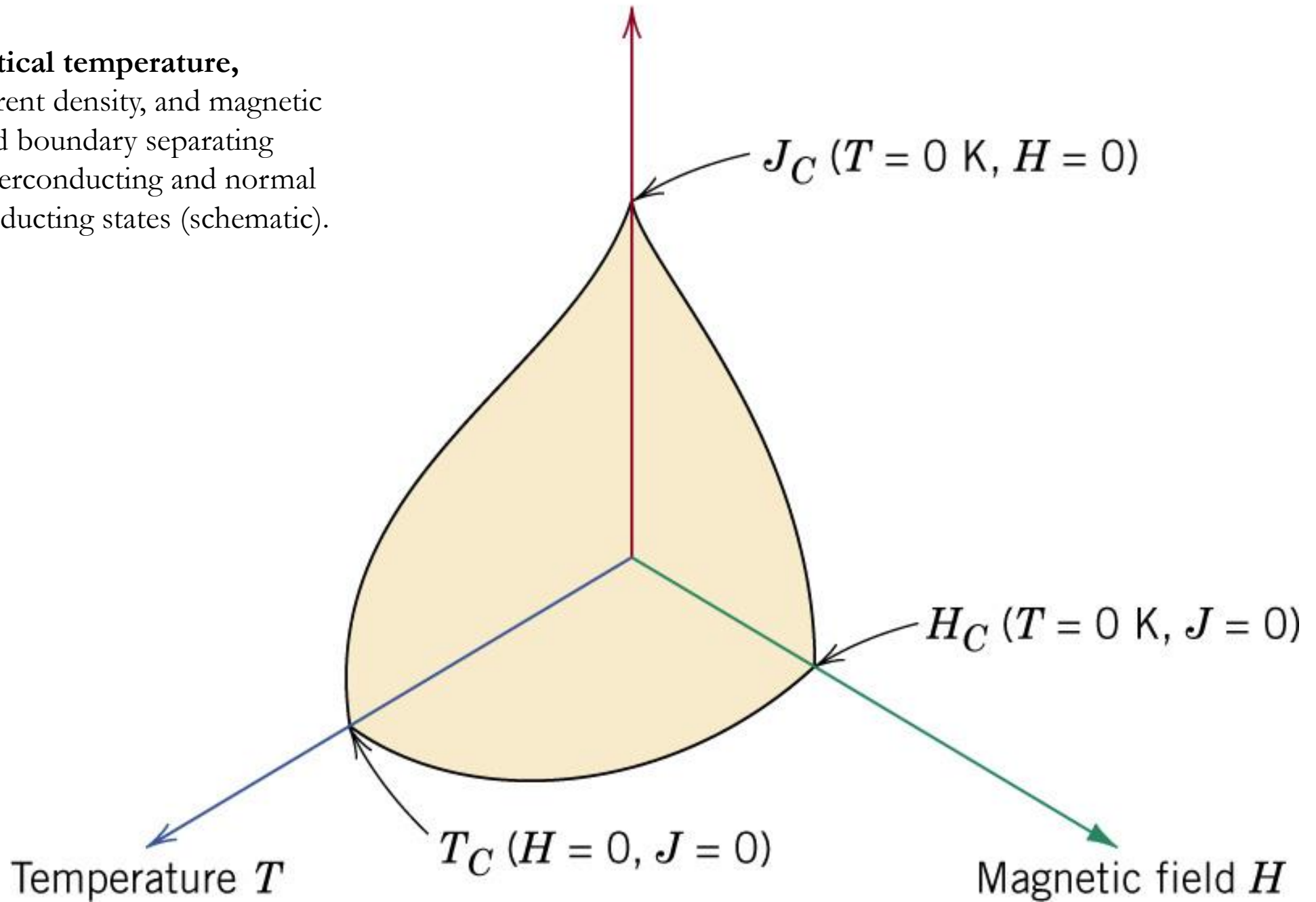


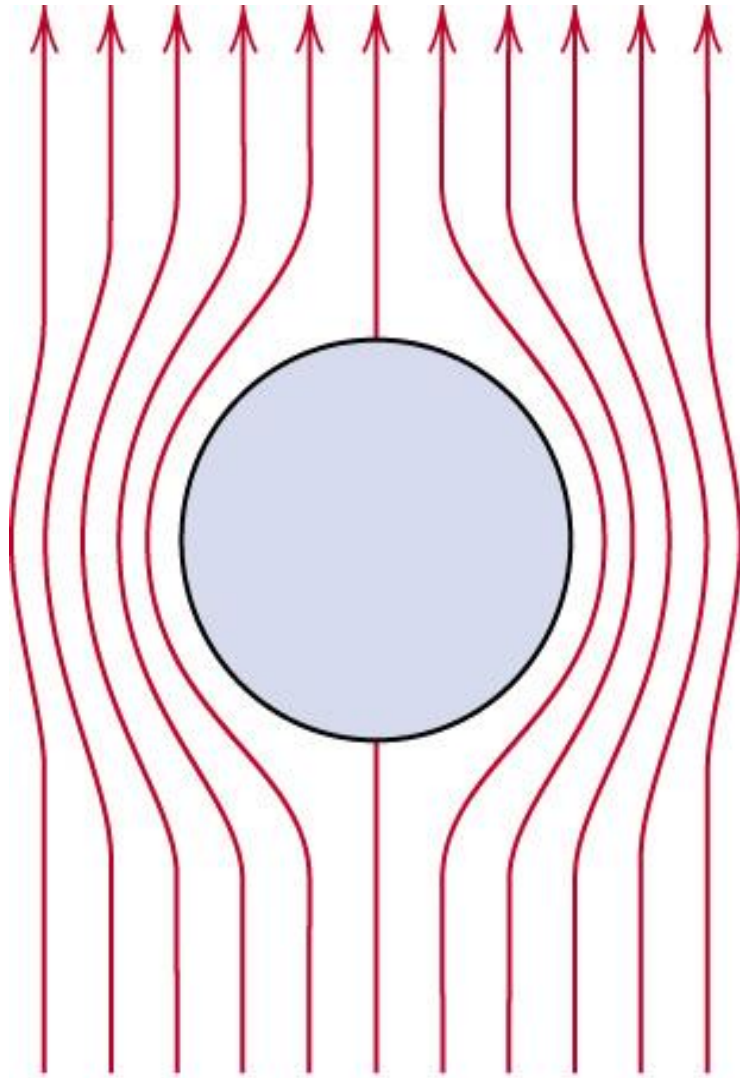
Superconductivity



Current density J

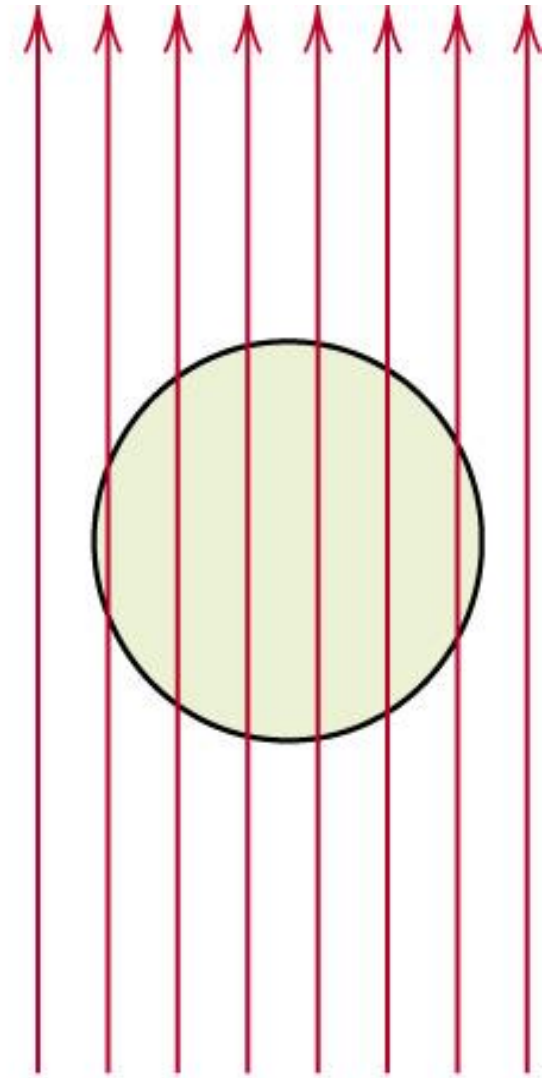
Critical temperature,
current density, and magnetic
field boundary separating
superconducting and normal
conducting states (schematic).





**Representation of
the Meissner effect.**

While in the superconducting state, a body of material (circle) excludes a magnetic field (arrows) from its interior.



The magnetic field penetrates the same body of material once it becomes normally conductive.

Critical temperature and Magnetic Fluxes for Selected Superconducting Materials

<i>Material</i>	<i>Critical Temperature T_C (K)</i>	<i>Critical Magnetic Flux Density B_C (tesla)^a</i>
Elements^b		
Tungsten	0.02	0.0001
Titanium	0.40	0.0056
Aluminum	1.18	0.0105
Tin	3.72	0.0305
Mercury (α)	4.15	0.0411
Lead	7.19	0.0803
Compounds and Alloys^b		
Nb–Ti alloy	10.2	12
Nb–Zr alloy	10.8	11
PbMo ₆ S ₈	14.0	45
V ₃ Ga	16.5	22
Nb ₃ Sn	18.3	22
Nb ₃ Al	18.9	32
Nb ₃ Ge	23.0	30
Ceramic Compounds		
YBa ₂ Cu ₃ O ₇	92	—
Bi ₂ Sr ₂ Ca ₂ Cu ₃ O ₁₀	110	—
Tl ₂ Ba ₂ Ca ₂ Cu ₃ O ₁₀	125	—
HgBa ₂ Ca ₂ Cu ₂ O ₈	153	—

^a The critical magnetic flux density ($\mu_0 H_C$) for the elements was measured at 0 K. For alloys and compounds, the flux is taken as $\mu_0 H_{C2}$ (in teslas), measured at 0 K.

^b **Source:** Adapted with permission from *Materials at Low Temperatures*, R. P. Reed and A. F. Clark (Editors), American Society for Metals, Metals Park, OH, 1983.

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FERRITES (铁氧体)

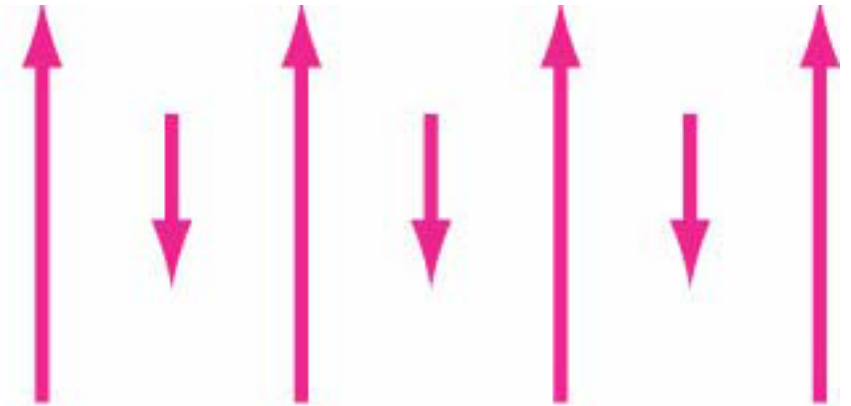
Ferrites are magnetic ceramic materials made by mixing iron oxide (Fe_2O_3) with other oxides and carbonates in the powdered form.

1. The powders are then pressed together and sintered at a high temperature.
2. Sometimes finishing machining is necessary to produce the desired part shape.
3. The magnetizations produced in the ferrites are large enough to be of commercial value, but their magnetic saturations are not as high as those for ferromagnetic materials. Ferrites have domain structures and hysteresis loops similar to those of ferromagnetic materials.
4. As in the case of the ferromagnetic materials, there are soft and hard ferrites.



Magnetically Soft Ferrites

Soft ferrite materials exhibit ferrimagnetic behavior. In soft ferrites there is a net magnetic moment due to two sets of unpaired inner-electron spin moments in opposite directions that do not cancel each other.



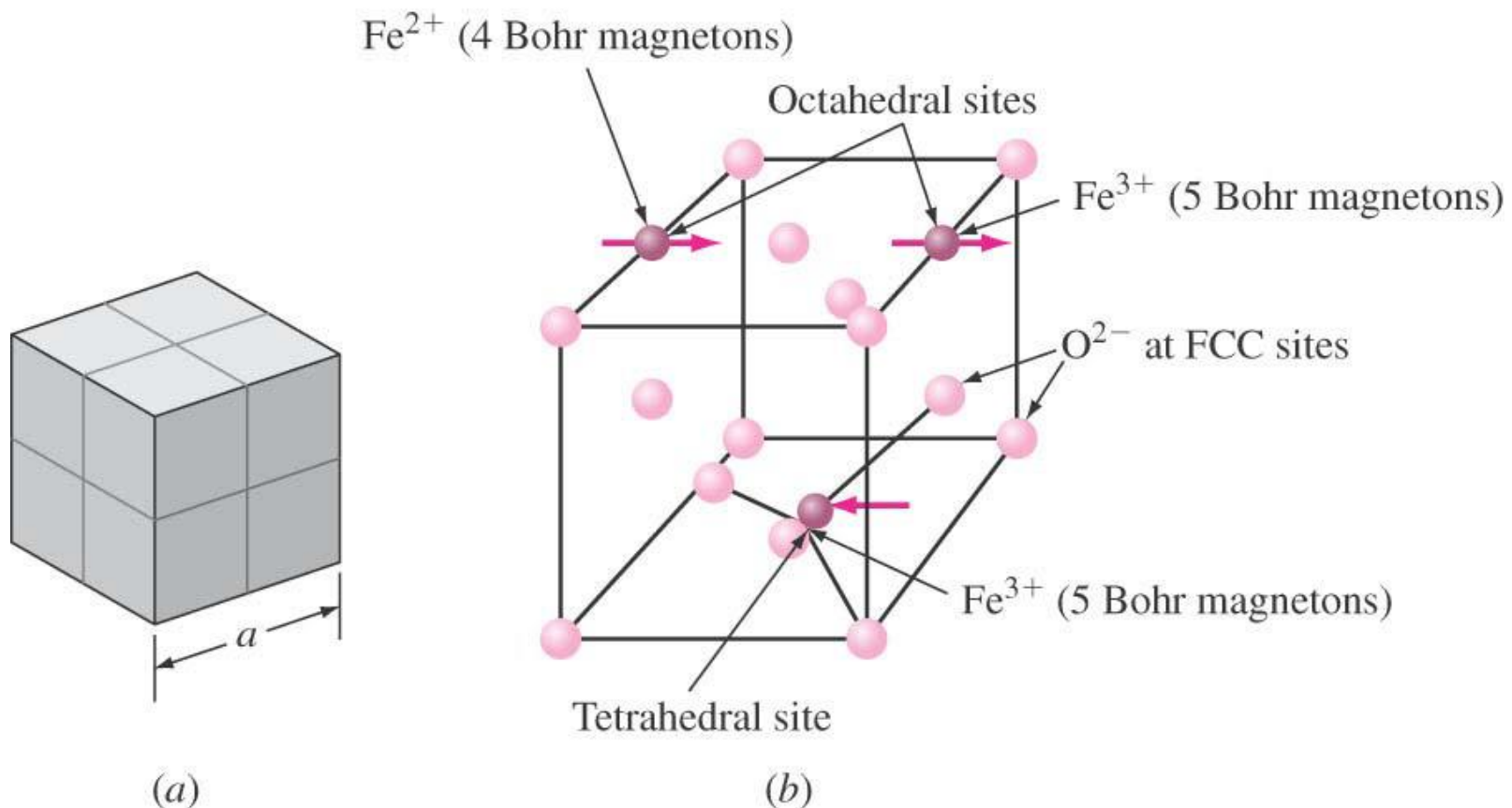
Composition and Structure of Cubic Soft Ferrites

Most cubic soft ferrites have the composition $\text{MO} \cdot \text{Fe}_2\text{O}_3$, where M is a divalent metal ion such as Fe^{2+} , Mn^{2+} , Ni^{2+} , or Zn^{2+} . The structure of the soft ferrites is based on the inverse spinel structure, which is a modification of the spinel structure of the mineral spinel ($\text{MgO} \cdot \text{Al}_2\text{O}_3$). Both the spinel and inverse spinel structures have cubic unit cells consisting of eight subcells. Each of the subcells consists of one molecule of $\text{MO} \cdot \text{Fe}_2\text{O}_3$. Since each subunit contains one $\text{MO} \cdot \text{Fe}_2\text{O}_3$ molecule and since there are seven ions in this molecule, each unit cell contains a total of $7 \text{ ions} \times 8 \text{ subcells} = 56 \text{ ions per unit cell}$.

Composition and Structure of Cubic Soft Ferrites

Each subunit cell has an FCC crystal structure made up of the four ions of the $\text{MO} \cdot \text{Fe}_2\text{O}_3$ molecule. The much smaller metal ions (M^{2+} and Fe^{3+}), which have ionic radii of about 0.07 to 0.08 nm, occupy interstitial spaces between the larger oxygen ions (ionic radius ~ 0.14 nm).

As previously discussed, in an FCC unit cell there are the equivalent of four octahedral and eight tetrahedral interstitial sites. In the normal spinel structure only half the octahedral sites are occupied, and so only 4 (8 subcells $\times$ 4 sites/subcell) = 16 octahedral sites are occupied. In the normal



SUMMARY

- A magnetic field can be produced by:
 - putting a current through a coil.
- Magnetic induction:
 - occurs when a material is subjected to a magnetic field.
 - is a change in magnetic moment from electrons.
- Types of material response to a field are:
 - ferri- or ferro-magnetic (large magnetic induction)
 - paramagnetic (poor magnetic induction)
 - diamagnetic (opposing magnetic moment)
- Hard magnets: large coercivity.
- Soft magnets: small coercivity.
- Magnetic storage media:
 - particulate γ -Fe₂O₃ in polymeric film (tape or floppy)
 - thin film CoPtCr or CoCrTa on glass disk (hard drive)

Summary I: Definition and Classification of Magnetic Materials

Parameter	Definition	Units/value	Conversion factor
H	Magnetic field strength	A/m	1 A/m = $4\pi \times 10^{-3}$ Oe (Oe)
H_c	Coercive field	A/m	
H_{cr}	Critical field	A/m	
M	Magnetization	A/m	
B	Magnetic flux density	T	T = Wb/m ² = kg/A ⁻¹ s ² = V/sm ² = 10 ⁴ G (G) 1 H = 1 J s ² C ⁻² 1 H/m = 1 Wbm ⁻¹ A ⁻¹
	Magnetic induction		
μ_0	Permeability of a vacuum	$4\pi \times 10^{-7}$ H/m	
μ	Permeability	H/m	
μ_r	Relative permeability	Dimensionless	0 K = -273°C
χ	Susceptibility	Dimensionless	
μ_{ion}	Net magnetic moment of an atom or ion	Am ²	
μ_s	Spin magnetic moment	Am ²	
μ_{orb}	Orbital magnetic moment	Am ²	0 K = -273°C
μ_B	Bohr magneton	9.274×10^{-24} Am ²	
θ_C	Curie temperature	K	
θ_N	Néel temperature	K	
T_c	Critical temperature for superconductivity	K	
C	Curie constant	K	

Class	Critical temperature	χ	Temperature variation of χ	Spontaneous magnetization	Structure on atomic scale
Diamagnetic	None	-10^{-6} to -10^{-5}	Constant	None	Atoms have no permanent dipole moments.
Paramagnetic	None	$+10^{-5}$ to $+10^{-3}$	$\chi = C/T$	None	Atoms have permanent dipole moments. Neighboring moments do not interact
Ferromagnetic	θ_C	Large (below θ_C)	Above θ_C , $\chi = C/(T - \theta)$, with $\theta \approx \theta_C$	Below θ_C , $M_s(T)/M_s(0)$ against T/θ_C follows a universal curve; above θ_C , none	Atoms have permanent dipole moments. Interaction produces parallel alignment
Antiferromagnetic	θ_N	As paramagnetic	Above θ_N , $\chi = C/(T \pm \theta)$, with $\theta \neq \theta_N$, below θ_N , χ decreases, anisotropic	None	Atoms have permanent dipole moments. Interaction produces antiparallel alignment
Ferrimagnetic	θ_C	As ferromagnetic	Above θ_C , $\chi \approx C/(T \pm \theta)$, with $\theta \neq \theta_C$	Below θ_C , does not follow universal curve; above θ_C , none	Atoms have permanent dipole moments. Interaction produces antiparallel alignment; moments not equal

Applications

Commercially prepared magnetic ceramics are used in a variety of permanent magnet:

Motors for electric toothbrushes and windshield wipers in Automobiles;

Transformer;

Speakers;

Telecommunications;

Information recording applications: ATM/Credit card.